Room temperature Mott transistor based on resistive switching in disordered V_2O_3 films grown on Si

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Electric field induced giant resistive switching triggered by insulator to metal transition (IMT) is one of the promising approaches for developing a new class of electronics often referred to as "Mottronics". Achieving this resistive switching by minimal external field at room temperature is of paramount research and technological interest. Mott-IMT is often associated with structural modification, which is very important for optoelectronic and actuator applications. Here, we report a giant resistive switching ~900 % at room temperature in disordered polycrystalline V_2O_3/Si thin film stabilized at IMT phase boundary and associated structural transformation under small electric field. Increase of electron population in a_{1g} band under field is responsible for Mott gap collapse that drives the structural transition. Furthermore, we also fabricated a room temperature Mott-FET with channel ON/OFF resistive ratio ~15. This study provides fundamental mechanism of the Mott-IMT in V_2O_3 as well as its device applications.

Realization of room temperature electric field controlled volatile resistive switching in silicon compatible functional materials is highly desirable for low power device applications in optoelectronics, neuromorphic computation, actuators, sensors, logic devices etc. [1,2,3,4]. Insulator to metal transition (IMT) triggered by electric field in Mott insulators is believed to be a very promising way to control the resistive state. Additionally, Mott materials have a fundamental edge over the conventional semiconductors on the speed, charge carrier density, power consumption and dimensionality scaling [5,6,7]. V_2O_3 is a classic Mott material with strong electron correlation, which exhibits a temperature driven metal to insulator transition (MIT) around 160 K [8,9]. The electric or optical field induced IMT in V_2O_3 is very fast, (~nsec) and requires low energy making it a promising material for high-speed, low-power electrical and optical switching and modulation applications [1,10,11,12,13].

In Mott insulators, strong electron correlations may couple to lattice and spin degrees of freedom. Crystallographic *c/a* ratio plays a crucial role in stabilizing the ground state of such a material. Isostructural paramagnetic metal (PM) to insulator (PI) transition in V_2O_3 has been reported by tuning the *c/a* ratio through chemical doping, disorder or strain [14,15,16]. Reduction of the *c/a* ratio increases the orbital overlap leading to increase in the separation between the Hubbard bands that opens up a Mott gap resulting in an insulating state. This iso-symmetric MIT provides huge opportunities for new electronic devices, where the ultimate aim is to stabilize the system at the metal-insulator phase boundary, such that a minimal external trigger leads to a maximal resistivity change. Several correlated oxide/ ferroelectric hetero-structures have been studied previously using V_2O_3, VO_2, Fe_2O_3, $LaNiO_3$ and $NdNiO_3$ [17,18,19,20,21,22 ,23], however, their

resistive switching is either less than 110% or at low temperatures. A resistive switching can also be achieved using electrolyte gating [24]. However, ionic-liquid modulation is not suitable for high-speed operation, device integration and in many cases induces an irreversible compositional change due to formation of oxygen vacancies at the liquid-solid interface [1,25]. Therefore, achieving high resistive switching by means of a small electric field at room temperature is of paramount research and technological interest.

To realize room temperature resistive switching, it is important to understand the fundamental mechanism of Mott IMT. In V_2O_3 it is still lacking. For example, Mott transition in many materials is accompanied by a structural phase transition (SPT). However, whether the crystal structure drives the Mott transition or electronic state drives SPT, is yet not clear. Here, with the help of simultaneous electric field dependent resistivity and Raman spectroscopy studies, we reveal that the electron-electron correlation is the key factor to drive the Mott transition in V_2O_3. Importantly, in this study the paramagnetic insulating phase (high resistive state -HRS) has been stabilized in polycrystalline V_2O_3/Si thin film at room temperature that can be transformed to the metallic state (low resistive state -LRS) by applying relatively small electric current (100mA) with a giant conductivity change ~ 900% without help of secondary drivers like ferroelectric interface. Further, for demonstrating its applications, we fabricated a prototype Mott Field Effect Transistor (MFET) with extremely low leakage current. This helped also to resolve the long standing fundamental concern whether the electric field induced IMT is of pure electronic origin or induced by joule heating.

Results

Thin film characterization

The microstructure of V_2O_3/Si thin film grown by Pulse laser deposition (PLD) is investigated using scanning transmission electron microscopy (STEM), X-ray diffraction (XRD), and Raman spectroscopy (details in supplementary note S1). Fig. 1a & b show the cross-sectional high angle annular dark field (HAADF) STEM image and corresponding elemental mapping. High resolution STEM image (Fig. 1c) depicts the highly disordered nano-crystallites. Fig. 1d represent the fast Fourier transformed pattern of the area marked by the blue boxes in Fig. 1c, which show the different orientations of the nano-crystallites. The crystallographic *c*/*a* ratio obtained from the atomically resolved STEM-HAADF image (Fig. 1e) is smaller than the bulk and V_2O_3/Al_2O_3 thin film (table 1). Fig. 1f shows the schematic of *a*-*c* plane of V_2O_3 unit cell. XRD and Raman studies shown in Fig. 1g, h further confirm the phase purity and reduction of *c*/*a* ratio in V_2O_3/Si thin film (details in note S1).

Table 1: Structural parameters of V_2O_3 thin film obtained from XRD and TEM compared with bulk [26].

Sample	*a* (Å)	*c*(Å)	*c/a*
Bulk	4.953	14.011	2.829
V_2O_3/ Al_2O_3	4.934+0.007	13.984+0.001	2.830+0.004
V_2O_3/Si (XRD)	4.974 $\pm$0.002 (0.4% tensile)	13.920$\pm$0.001 (0.6% compressive)	2.798$\pm$0.001
V_2O_3/Si (TEM)	5.10	14.28	2.80

Temperature dependence of resistance under different electric field

Fig. 1i shows the resistance (R) vs temperature (T) behavior measured on the V_2O_3/Si thin film using linear four probe method under application of different electric fields (currents). A huge increase of resistance with decreasing T is observed under small probe current (10μA). The negative slope (dR/dT) indicates the insulating nature of the thin film even at room temperature in sharp contrast to the room temperature metallic phase in bulk and epitaxial V_2O_3/Al_2O_3 thin film (shown in Fig. S1) [27, 28]. Interestingly, with increase in the probe current, the change in resistance as a function of T becomes smaller and smaller (Details in note S2). The present results show that the room temperature insulating phase and resistive switching can be realized in undoped polycrystalline thin films with structural defects playing the role of Cr-doping [14].

Theoretical studies predicted reduction in Mott gap with a metastable state due to phase-coexistence and through electron injection collapses the gap under a very small applied electric field [29]. Defects and disorder in Mott materials often facilitate such phenomenon [11]. To check the possibility of a change in the electronic state driven by the electric field as well as the effect of electric field on associated crystal structure in the present disordered polycrystalline V_2O_3/Si thin film, we performed *in-situ* Raman and voltage (V)-current (I) characteristics at room temperature and low temperatures.

Electric field induced resistance switching and crystal structure change at 300K and 7K

A schematic of the *in-situ* measurements geometry is shown in Fig. 2a. Fig. 2b presents V-I characteristics at room temperature. Non-linearity in V-I characteristics indicate the electric filed induced resistive switching in V_2O_3/Si film. This is in sharp contrast to the metallic V_2O_3/Al_2O_3 film which shows linear V-I response at room temperature (shown in Fig. S1b). Corresponding differential conductance (DC) is plotted in the Fig. 2b. DC is obtained as DC = $\frac{\sigma_d(\mathrm{I}) - \sigma_d(0)}{\sigma_d(0)} \times 100\%$. , where σ_d(I) is conductance, defined as σ_d(I)= $\left(\frac{dI}{dV}\right)_{I=I}$ and σ_d(0)= $\left(\frac{dI}{dV}\right)_{I=0}$. It shows a high DC change ~ 900% at room temperature on the application of small current (100mA). On the other hand, the metallic V_2O_3/Al_2O_3 thin film does not show any DC as shown in Fig. S1c. To probe the associated structural modification, we performed the Raman spectroscopy measurements. Fig. 2c shows the Raman spectra collected under different current values at 300K. Position of the A_{1g} and E_g Raman modes under zero field (254 and 236 cm^{-1}, respectively) are at higher wave numbers than observed in the metallic bulk and V_2O_3/Al_2O_3 thin film (242 and 210 cm^{-1}) [30] and is attributed to the lower *c/a* ratio in the V_2O_3/Si thin film [31]. For comparison, the Raman spectrum of metallic V_2O_3/Al_2O_3 thin film is also plotted in Fig 2c. The position of both the modes shows a shift towards the lower wave number and approaches to the metallic V_2O_3/Al_2O_3 film with increasing current. Thus, the Raman study indicates that on application of electric field the insulating R′ phase transforms to the metallic R phase. Therefore, these results demonstrate a unique way to tune the crystallographic unit cell by application of electric current which is the central factor for optoelectronic applications [1,2,32,33,34]. As discussed before, the V_2O_3/Si film is a room temperature paramagnetic insulator (HRS) having corundum structure (R′ phase) with lattice parameters slightly differing from the metallic state (LRS) corundum structure (R phase). With the application of electric field, the phonon modes show a softening without change in symmetry or undergoes isostructural HRS (R′) to LRS (R) phase transition originated from the strong electron-electron correlation (other possibilities are discarded in note S2).

Similarly, to probe the electric field effect on low temperature electronic and structural phase, we have performed in-situ Raman at 7K. With lowering T, the room temperature insulating R′ phase transforms to